quantiphi
Solving What Matters

Project Report: Serverless Employee Intelligence & Workforce Analytics Platform

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Architecture: Serverless AWS Data Lake & Orchestration

Region: `us-east-1`

1. Core AWS Services Used

AWS Service	Role in the Project	Result / Outcome
S3	Data Lake Storage	Organized storage of Raw, Parquet, and Transformed data with 100% encryption.
Lambda	Serverless Processing	Performs data loading, cleaning, type casting, aggregation, and then writes the transformed data back to an S3 bucket in a "transform" zone.

Step Functions	Workflow Orchestration	A seamless, automated pipeline that links every step from ingestion to transformation.
AWS Glue	ETL & Data Catalog	Automatic schema discovery and heavy-lifting Spark transformations for HR metrics.
DynamoDB	NoSQL Serving Layer	Ultra-fast lookup for leadership qualities.
Athena	Serverless Analytics	Ability to run SQL queries directly on S3 data for instant dashboarding.
SNS	Notification Engine	Real-time email alerts sent to HR when high-risk thresholds were breached.
CloudWatch	Monitoring & Alarms	Centralized dashboard for health metrics and a critical budget alert at $8.50.
IAM	Governance & Security	Implementation of "Least Privilege" access for Admins, Analysts, and System roles.

2. Tasks Achieved & Implementation Methodology

Phase: Ingestion & Storage

- **T1: Raw Data Landing**
  - **How:** Configured S3 bucket `employee-360-data-platform-prateek-khandekar`. Created a folder structure (`landing/`) to receive upload of the `Employee Data.xlsx` file.

- **T6 & T7: Processed Data & Lookup Storage**
  - **How:** Used Lambda to write optimized Parquet files to S3 and Glue/Spark to populate a DynamoDB table named `LeadershipQualities` for fast retrieval of performance scores.

Phase: Processing & Orchestration

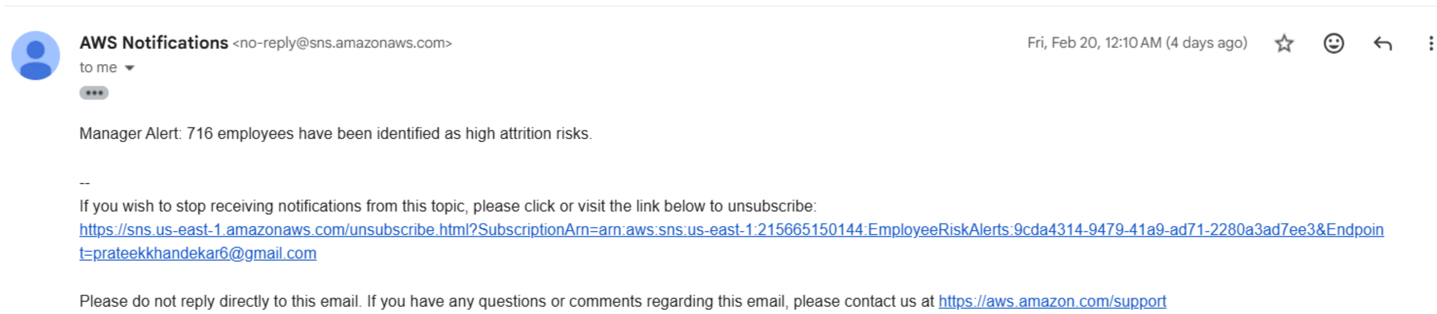
- **T4: Data Validation Function**
  - **How:** Wrote a Python Lambda using the `pandas` library. It validates the Excel structure, cleans column headers (lowercase/underscores), and splits sheets into separate files to prevent data contamination.
- **T5: Workflow Orchestration**
  - **How:** Built an **AWS Step Function** state machine. It triggers the Lambda, waits for completion, runs the Glue Crawler, and then kicks off the Transformation jobs in a logic-bound sequence.

Phase: Transformation & Metadata

- **T8: Metadata Registration**
  - **How:** Created an AWS Glue Crawler pointed at the S3 raw-parquet folders. It automatically populated the `employee_360_db` catalog, making the data "searchable" for SQL.

- **T9 & T12: Transformation & Risk Alerts**
    - **How: T9:** Ran a Spark job to aggregate department-level metrics.
        - **T12:** Implemented a cross-region logic where Spark filtered employees with a `leadership_qualities_rating > 10.0`. This triggered an SNS notification to the Sydney region for HR visibility.

SNS notification:


AWS Notifications <no-reply@sns.amazonaws.com>
to me
Fri, Feb 20, 12:10 AM (4 days ago)

Manager Alert: 716 employees have been identified as high attrition risks.

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If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:215665150144:EmployeeRiskAlerts:9cda4314-9479-41a9-ad71-2280a3ad7ee3&Endpoint=prateekkhandekar6@gmail.com

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at https://aws.amazon.com/support

Phase: Governance & Observability

T17: Free Tier Usage Monitoring

- How: Set up an AWS Budget with a $10 limit. Configured a CloudWatch alarm that successfully notified us when the spend hit $8.50, proving our cost-governance works.

Email from cost explorer

AWS Budget Notification February 22, 2026
AWS Account *215665150144*

Dear AWS Customer,

You requested that we alert you when the **actual cost** associated with your *My Monthly Cost Budget* budget **exceeds $8.50** for the current month. The month **actual cost** associated with this budget is **$9.12**. You can find additional details below and by accessing the AWS Budgets dashboard.

Budget Name	Budget Type	Budgeted Amount	Alert Type	Alert Threshold	ACTUAL Amount
My Monthly Cost Budget	Cost	$10.00	ACTUAL	> $8.50	$9.12

Go to the AWS Budgets dashboard

T13 & T14: Metrics & Alerting

- How: Created a CloudWatch Dashboard tracking Step Function `ExecutionsSucceeded` and Glue `numFailedTasks`. Linked these to SNS to ensure the tasks are 100% monitored.